OXFORD
INTERNATIONAL
AQA EXAMINATIONS

Please write clearly in block capitals.

Centre number						Candidate number				

Surname ______________________________

Forename(s) ______________________________

Candidate signature ______________________________
I declare this is my own work.

INTERNATIONAL GCSE
BIOLOGY
Paper 1

Wednesday 27 October 2021 07:00 GMT Time allowed: 1 hour 30 minutes

Materials
For this paper you must have:
- a ruler with millimetre measurements
- a scientific calculator.

Instructions
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice
In all calculations, show clearly how you work out your answer.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	

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Answer **all** questions in the spaces provided.

0 1 The human nervous system helps humans react to their surroundings.

0 1 . 1 Receptors are part of the nervous system.

Name the organ where each type of receptor is found.

[3 marks]

Taste (chemical) receptor ______________________________

Light receptor ______________________________

Sound receptor ______________________________

0 1 . 2 Reflex actions are automatic responses to stimuli.

Reflex actions follow a pathway.

Table 1 shows the knee jerk reflex action.

Complete **Table 1**.

[3 marks]

Choose answers from the box.

Coordinator	**Effector**	**Response**	**Stimulus**

Table 1

Direction of pathway	Part of reflex pathway	In the knee jerk reflex action
↓		Tap the knee
	Receptor	Knee tendon
	Sensory neurone	Leg neurone A
		Spinal cord
	Motor neurone	Leg neurone B
		Leg muscle
		Leg kicks out

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0 1 . 3 A synapse is a gap between two neurones.

Describe how the nerve impulse is transferred across a synapse to continue in the second neurone.

[2 marks]

8

Turn over for the next question

0 2 Animals have adaptations which help them survive in the conditions where they normally live.

Figure 1 shows a cat flea.

Cat fleas normally live in the fur of a cat.

Figure 1

0 2 . 1 Draw **one** line from each adaptation to how the adaptation helps the flea survive.

[3 marks]

Adaptation	**How the adaptation helps the flea survive**
Flattened body	Can bite to feed on cat blood
Hard outer case	Can grip cat fur so it holds on firmly
Hooks on ends of legs	Can move easily through the fur
Sharp mouth parts	Cannot be easily damaged

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0 2 . 2 Animal adaptations can be behavioural, functional or structural.

What type of adaptations are shown in **Figure 1**?

Tick (✓) **one** box.

[1 mark]

Behavioural ☐

Functional ☐

Structural ☐

0 2 . 3 Complete the sentences to describe the relationship between the flea and the cat.

[2 marks]

The flea lives on the cat and benefits from the cat.

The flea is a ______________________________.

The cat is harmed by the flea.

The cat is the __________________________ to the flea.

0 2 . 4 The flea takes blood from the cat for food.

A digestive enzyme breaks down the protein haemoglobin in the cat blood.

Name the type of product from haemoglobin digestion.

[1 mark]

____

0 2 . 5 Tardigrades are very small animals.

They are adapted:

- to survive temperatures as low as -100 °C
- to resist drying out.

What is the scientific term for organisms that can survive in these conditions?

[1 mark]

____

8

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0 3 This question is about blood.

0 3 . 1 Complete the sentence.

Choose the answer from the box.

[1 mark]

plasma	**urine**	**water**

The liquid part of the blood is called ________________________.

0 3 . 2 Complete the sentence.

Choose the answer from the box.

[1 mark]

liver cells	**skin cells**	**white blood cells**

Cells suspended in the liquid part of the blood include red blood cells and

__________________________.

0 3 . 3 Complete the sentence.

Choose the answer from the box.

[1 mark]

glucose	**nitrogen**	**urea**

The waste substance transported by the blood from the liver to the kidneys

is ____________________ .

Do not write outside the box

0 3 . 4 Complete the sentence.

Choose the answer from the box.

[1 mark]

carbon dioxide	**glycogen**	**nitrogen**

The waste substance transported in the blood from the body organs to the lungs is ________________________________.

0 3 . 5 What is the function of the red blood cells?

Tick (✓) **one** box.

[1 mark]

To prevent backflow of blood ☐

To produce antibodies ☐

To transport oxygen to the organs ☐

Question 3 continues on the next page

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0 3 . 6 A boy falls over and cuts his knee.

Explain the steps which take place to form a clot over the cut.

[5 marks]

10

Do not write outside the box

0 4 Students in a school investigated the effect of the level of exercise on heart rate.

This is the method used.

1. Select six volunteers randomly from the school.
2. Ask volunteers to sit on a low bench for 10 minutes.
3. Measure and record the heart rate of each volunteer.
4. Ask each volunteer to step onto and down from the bench 5 times.
5. Measure and record their heart rate.
6. Allow volunteers to rest.
7. Repeat steps 4 to 6 using 10, 15 and then 20 step-ups.
8. Calculate a mean heart rate for the six volunteers at each level of exercise.

0 4 . 1 Give the dependent variable in this investigation.

[1 mark]

____

0 4 . 2 Give **two** control variables the students should have used in their investigation.

[2 marks]

1 ____

____

2 ____

____

0 4 . 3 The students could not measure heart rate using electronic monitors.

Describe another method the students could use to measure heart rate.

[2 marks]

____

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Turn over ►

Table 2 shows the results.

Table 2

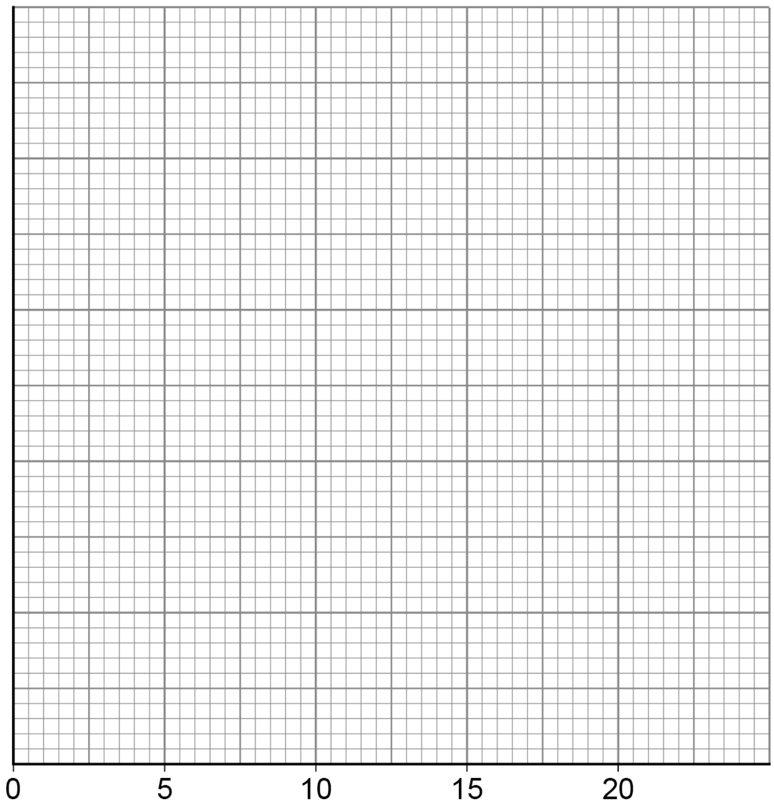
Number of step-ups	Mean heart rate in beats per minute
0 (resting)	98
5	143
10	161
15	174
20	174

0 4 . 4 Plot the data from **Table 2** on **Figure 2**.

You should:
- label both the axes
- provide a suitable scale for the y-axis
- plot the data in **Table 2** on **Figure 2**
- draw a line of best fit.

[5 marks]

Figure 2



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0 4 . 5 Give the conclusion that can be made about the effect of the level of exercise on heart rate.

Use data from **Table 2.**

[2 marks]

Question 4 continues on the next page

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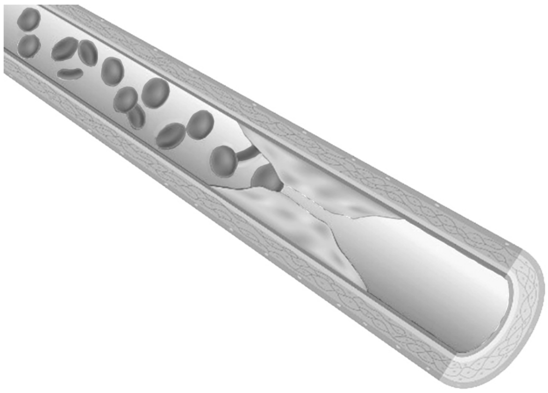
Figure 3 shows the inside of the coronary artery of a person.

Figure 3



0 4 . 6 Explain why the person is likely to suffer from coronary heart disease.

[3 marks]

0 4 . 7 Name the surgical device doctors could insert into the coronary artery to keep it open.

[1 mark]

16

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0 5 A woman gave birth to twin boys.

The twins were formed from **one** egg that was fertilised by **one** sperm.

The resulting embryo split into two at a very early stage and developed into identical twins.

Both twins had brown eyes.

The twins had different birth weights.

0 5 . 1 At birth:

- Twin **A** was 2.5 kg.
- Twin **B** was 3.1 kg.

Calculate how much **more** twin **B** weighed compared with twin **A**.

Give your answer as a percentage.

[3 marks]

Percentage = ____________________________ %

Do not write outside the box

0 5 . 2 Variation describes differences in the characteristics of organisms of the same species.

Complete **Table 3** for the characteristics of the twins.

Place **two** ticks (✓) in **each** column.

One tick each for:

- the cause of the characteristic
- the type of variation shown.

The first column has been completed.

[4 marks]

Table 3

		Characteristic		
		Male sex	**Brown eyes**	**Birth weight**
Cause of characteristic	**Genetic only**	✓		
	Environmental only			
	Genetic and environmental			
Type of variation	**Continuous**			
	Discontinuous	✓		

Do not write outside the box

0 5 . 3 The father of the twins has brown eyes.

Explain how the characteristic of brown eyes has been passed from the father to both of his children.

[4 marks]

0 5 . 4 The cells in a human embryo are stem cells.

The stem cells change into many types of cells with specific functions.

What name is given to this process?

[1 mark]

Turn over ►

Therapeutic cloning is a technique which can be used to produce stem cells to treat a patient suffering from a condition such as paralysis.

In therapeutic cloning:

- the nucleus is removed from a donor egg cell
- a nucleus from a body cell from the patient is placed in the donor egg cell
- the donor egg cell develops into an embryo in a laboratory
- when the embryo is 5 days old some cells are removed and cultured until there are enough cells to treat the patient
- the cells are injected into the patient where they repair the damaged tissue
- the remaining embryo cells are not used.

0 5 . 5 One advantage of therapeutic cloning is that it could cure paralysis.

Explain **one** other advantage and **one** disadvantage of using stem cells produced by therapeutic cloning to treat paralysis.

[4 marks]

Advantage ____________________

Disadvantage ____________________

0 5 . 6 Some adult tissues contain stem cells.

Name **one adult** human tissue which contains stem cells.

[1 mark]

Do not write outside the box

0 5 . 7 Scientists can now take adult human stem cells and turn them into special stem cells called iPSCs.

iPSCs can specialise into almost any type of human body cell.

In 2018 scientists turned iPSCs into pancreas cells.

Explain how pancreas cells from iPSCs could be used to treat a patient with Type 1 diabetes.

[3 marks]

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20

Turn over for the next question

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0 6 Particles are transported into and out of cells by different processes.

0 6 . 1 Draw **one** line from each transport process to the definition for that process.

[2 marks]

Transport process	**Definition**
	The movement of water from a concentrated solution to a dilute solution through a partially permeable membrane
Active transport	
	The movement of water from a dilute solution to a concentrated solution through a partially permeable membrane
	The movement of particles from a low concentration to a high concentration requiring energy
Osmosis	
	The spreading out of particles when concentrations are equal

0 6 . 2 All cells respire aerobically.

Aerobic respiration requires oxygen.

Name the transport process that moves oxygen into cells.

[1 mark]

____

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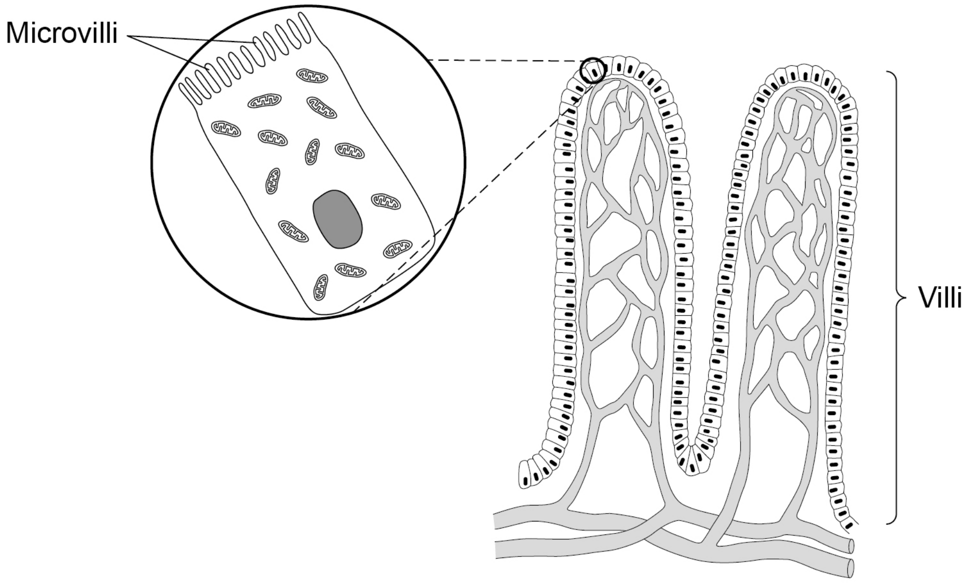
0 6 . 3 **Figure 4** shows part of the lining of the small intestine.

Figure 4



Food molecules are absorbed from the small intestine into the blood in the capillaries.

Describe how **four** features of the small intestine increase the efficiency of absorption.

Use **Figure 4** to help you.

[4 marks]

1 ____

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2 ____

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3 ____

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4 ____

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Turn over ►

Scientists analysed the fluid in the small intestine and the blood of a woman.

They tested the fluid and the blood:

- after fasting (not eating) for 12 hours
- 30 minutes after eating a meal.

The meal contained mainly proteins and carbohydrates such as starch.

Table 4 shows the data.

Table 4

Food molecules	**Concentration in arbitrary units**			
	Fluid in small intestine after fasting	**Fluid in small intestine 30 minutes after meal**	**Blood after fasting**	**Blood 30 minutes after meal**
Amino acids	570	9500	1250	1700
Glucose	1.2	52.7	4.6	6.1
Starch	0.3	10.1	0	0

0 6 . 4 Explain how the proteins and the carbohydrates are:

- broken down
- transported into the blood after a meal
- transported into the blood after fasting for 12 hours.

Use data from **Table 4** to support your answer.

[6 marks]

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0 6 . 5 The woman added a lot of salt to her meal.

The concentration of ions in her blood increased.

Explain how her body reacted to the large salt intake.

[2 marks]

15

Turn over for the next question

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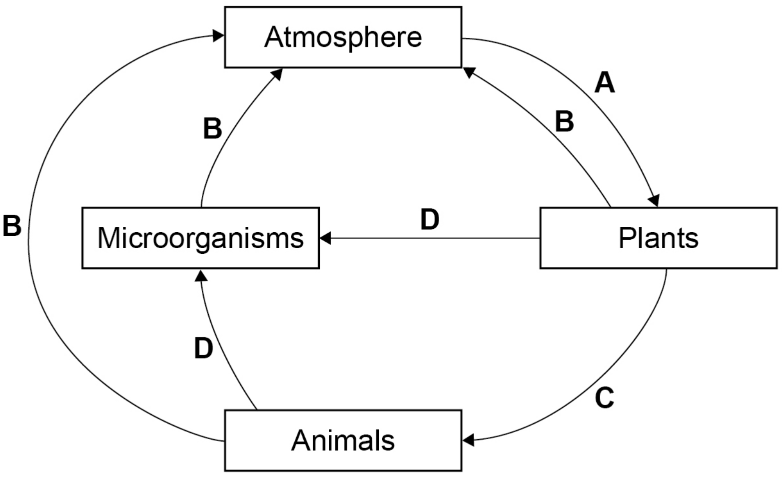
0 7 **Figure 5** shows part of the carbon cycle.

Figure 5



0 7 . 1 Name process **C** on **Figure 5**.

[1 mark]

____

0 7 . 2 Write the chemical equation for process **A** shown on **Figure 5**.

The chemical formula for glucose is $C_6H_{12}O_6$.

[2 marks]

____________ + ____________ → ____________ + ____________

0 7 . 3 Write the chemical equation for process **B** shown on **Figure 5**.

The chemical formula for glucose is $C_6H_{12}O_6$.

[2 marks]

____________ + ____________ → ____________ + ____________

Question 7 continues on the next page

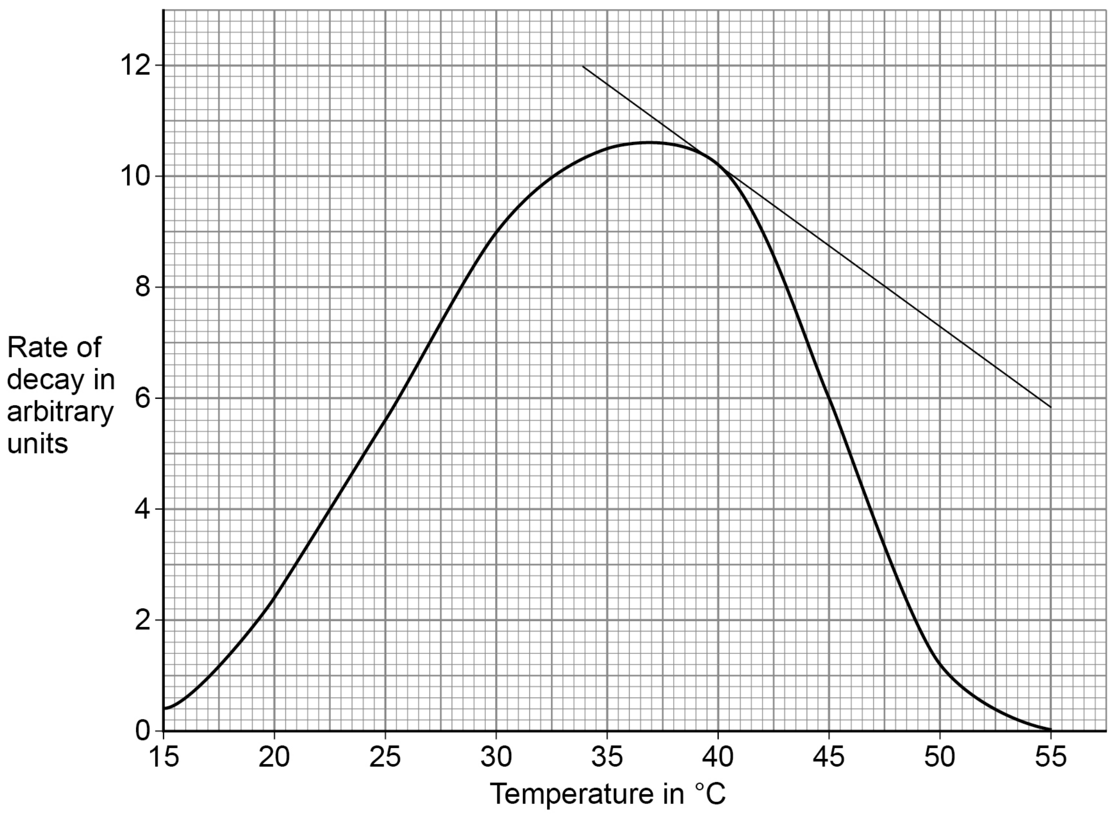
Turn over ►

Decay is caused by microorganisms.

Decay happens fastest when conditions are warm and moist and there is a good supply of oxygen.

Figure 6 shows how the rate of decay in garden waste changes with temperature.

Figure 6



0 7 . 4 Determine the decrease in the rate of decay from 40 °C to 45 °C.

[2 marks]

____

____

____

Decrease = ____________________ arbitrary units

Do not write outside the box

0 7 . 5 Determine the **rate** of decrease at 40 °C.

Use the tangent in **Figure 6**.

[3 marks]

Rate of decrease = ____________________ arbitrary units per °C

0 7 . 6 Explain the shape of the graph in **Figure 6**.

[2 marks]

0 7 . 7 Processes add and remove carbon and other materials from a community as part of a constant cycle.

What needs to happen to these processes to create a stable community?

[1 mark]

13

END OF QUESTIONS

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